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Chapter 1

Networking Devices, Technologies, and Models

Although the CCNA certification exams test you on some difficult and in-depth networking issues, the ICND1 exam also tests you on basic networking knowledge. For the ICND1 exam, you are expected to know the purpose of different network services, devices, cabling, and the OSI model. This chapter helps you prepare for those questions by testing you on basic networking concepts.

The Questions You'll Work On

In this chapter, you'll review questions concerning the following topics:

- Recognizing the purpose of networking services such as DHCP, DNS, NAT, and authentication services
- Understanding the difference between hubs, bridges, switches, and routers
- Understanding the OSI model and what devices and protocols run at each layer of the OSI model
- Knowing when to use a straight-through cable or a crossover cable
- Understanding the difference between a collision domain and a broadcast domain

What to Watch Out For

Don't let common mistakes trip you up; watch for the following when working with these questions:

- Straight-through cables are used to connect dissimilar devices, whereas crossover cables are used to connect similar devices. The one gotcha on this is that a crossover cable is used to connect a workstation to a router.
- Know the OSI model and what devices and protocols run at the different layers. Ensure that you know the difference between a layer-2 address and a layer-3 address.
- Hubs are layer-1 devices, whereas switches and bridges are layer-2 devices. A router is a layer-3 device.
- Web servers deliver HTML pages using HTTP as a delivery protocol. FTP is the Internet protocol for downloading files, SMTP is the Internet protocol for delivering e-mail, and POP3 and IMAP are Internet protocols for reading e-mail. An authentication server is responsible for verifying usernames and passwords before someone can gain network access.

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Network Services

1–7 Choose the best answer(s).

1. Your manager asks you which service is responsible for translating the source IP address of a packet to the IP address of the public interface on the router.

(A) DHCP

(B) NAT

(C) DNS

(D) HTTP

2. You are troubleshooting a communication problem. You seem to be able to communicate with Glen's website by IP address, but not by the fully qualified domain name (`www.gleneclarke.com`). What is most likely the problem?

(A) DHCP

(B) NAT

(C) DNS

(D) HTTP

3. Which network service can be configured on your router that is responsible for assigning IP addresses to systems on the network?

(A) DHCP

(B) NAT

(C) DNS

(D) HTTP

4. What service on the network is responsible for converting the FQDN to an IP address?

(A) Authentication server

(B) DNS server

(C) DHCP server

(D) NAT

5. What service on the network is responsible for verifying username and passwords when the user attempts to log on?

(A) Authentication server

(B) DNS server

(C) DHCP server

(D) NAT

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6. You are monitoring network traffic and you notice a number of DHCP discover messages on the network. Which of the following is the destination address of the DHCP discover message?

(A) 0F-1B-3C-2F-3C-2A

(B) FF-FF-FF-FF-FF-FF

(C) 192.168.4.5

(D) 192.168.0.0

7. When a client system boots up and requests an IP address, it first must send out which message?

(A) DHCP request

(B) DHCP discover

(C) DHCP response

(D) DHCP ACK

Network Devices and Communication

8–17 Choose the best answer(s).

8. What device is responsible for regenerating the signal so that the signal can travel a greater distance?

(A) Bridge

(B) Router

(C) Repeater

(D) Switch

(E) Hub

9. Which device filters traffic by looking at the destination address of the frame and then forwards the frame to the port that the destination system resides on?

(A) Hub

(B) Router

(C) Repeater

(D) Switch

10. Which of the following is a layer-3 device?

(A) Bridge

(B) Router

(C) Repeater

(D) Switch

(E) Hub

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11. A device that can send and receive information, but not at the same time, is said to be _____.

(A) Simplex

(B) Full duplex

(C) Multicast

(D) Half-duplex

12. A message that is sent out on the network and is destined for all systems is known as a _____ message.

(A) Unicast

(B) Multicast

(C) Full duplex

(D) Broadcast

13. A message that is sent out on the network and is destined for a group of systems is known as a _____ message.

(A) Unicast

(B) Multicast

(C) Full duplex

(D) Broadcast

14. A group of systems that can receive one another's broadcast messages is known as a _____.

(A) Collision domain

(B) Active directory domain

(C) Fully qualified domain name

(D) Broadcast domain

15. You are monitoring network traffic and notice that there is a large number of broadcast messages sent across the wire. You would like to separate your network into multiple broadcast domains. How can you do this? (Select two.)

(A) Switch

(B) VLANs

(C) Router

(D) Bridge

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16. A group of systems that can have their data collide with one another is known as a _____.

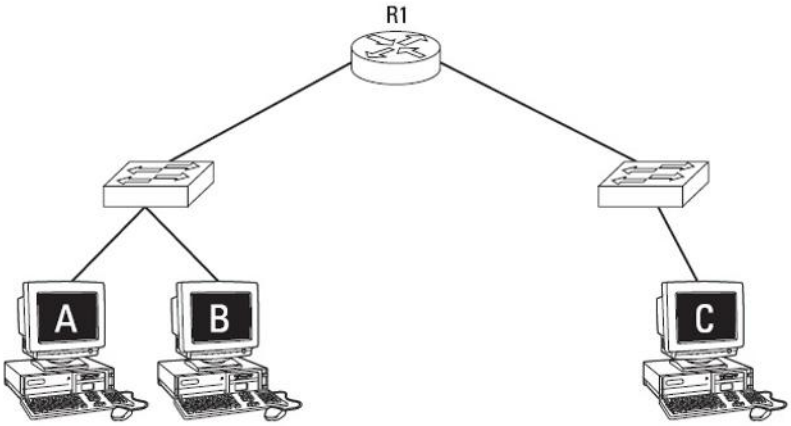
(A) Broadcast domain

(B) VLAN

(C) Collision domain

(D) Multicast

17. How many broadcast domains and collision domains are there in the diagram below?



(A) 1 broadcast domain and 5 collision domains

(B) 2 broadcast domains and 3 collision domains

(C) 1 broadcast domain and 3 collision domains

(D) 2 broadcast domains and 5 collision domains

OSI Model and Network Standards

18–31 Choose the best answer(s).

18. Sue is having trouble understanding some network concepts and asks you to help identify address types. Which of the following is considered a layer-2 address?

(A) 192.168.2.200

(B) `www.gleneclarke.com`

(C) COMPUTER1

(D) 00-AB-0F-2B-3C-4E

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19. You are troubleshooting communication to a network by looking at the link light on the switch. What layer of the OSI model are you troubleshooting when looking at a link light?

(A) Application

(B) Physical

(C) Network

(D) Data link

20. What layer of the OSI model is responsible for breaking the data into smaller segments?

(A) Data link

(B) Physical

(C) Network

(D) Transport

21. Which of the following is considered a layer-3 address?

(A) 192.168.2.200

(B) `www.gleneclarke.com`

(C) COMPUTER1

(D) 00-AB-0F-2B-3C-4E

22. What layer of the OSI model is responsible for routing and logical addressing?

(A) Network

(B) Physical

(C) Data link

(D) Transport

23. Which of the following are considered layer-2 devices? (Choose two.)

(A) Bridge

(B) Router

(C) Repeater

(D) Switch

(E) Hub

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24. Which of the following are considered layer-1 devices? (Choose two.)

(A) Bridge

(B) Router

(C) Repeater

(D) Switch

(E) Hub

25. What is the application layer protocol for delivering e-mail across the Internet?

(A) FTP

(B) SNMP

(C) SMTP

(D) POP3

26. Which layer of the OSI model works with packets?

(A) Layer 1

(B) Layer 2

(C) Layer 3

(D) Layer 4

27. What is the application layer protocol for receiving e-mail over the Internet?

(A) FTP

(B) SMTP

(C) SNMP

(D) POP3

28. Which Gigabit Ethernet standard uses UTP cabling to reach 1000 Mbps?

(A) 1000BaseTX

(B) 1000BaseSX

(C) 1000BaseCX

(D) 1000BaseLX

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29. Which 10 Gigabit Ethernet standard uses multimode fiber-optic cabling?

(A) 10GBaseLR

(B) 10GBaseER

(C) 10GBaseSR

(D) 1000BaseSX

30. Which of the following addresses does a router use to determine where a packet needs to be delivered? (Choose two.)

(A) 24.56.78.10

(B) 00-3B-4C-2B-00-AF

(C) A layer-3 address

(D) A layer-2 address

31. Which layer of the OSI model works with frames?

(A) Layer 1

(B) Layer 2

(C) Layer 3

(D) Layer 4

Network Cabling

32–38 Choose the best answer(s).

32. What type of cable would you use if you wanted to connect a system to an RJ-45 port on a switch?

(A) Fiber

(B) Crossover

(C) Straight-through

(D) Thinnet

33. You wish to network two systems by connecting a computer directly to another computer. Which type of cable would you use?

(A) Fiber

(B) Crossover

(C) Straight-through

(D) Thinnet

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34. You need to create a crossover cable. What wires would you cross on one of the ends?

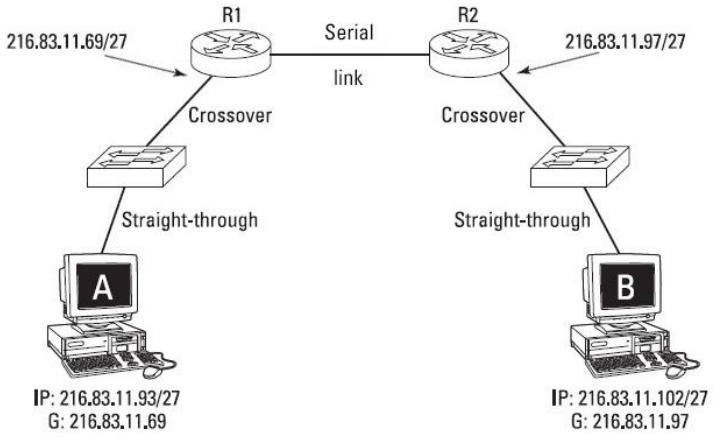
(A) 1 and 2 with 3 and 4

(B) 2 and 4 with 6 and 8

(C) 2 and 4 with 5 and 6

(D) 1 and 2 with 3 and 6

35. See figure below. You are trying to ping computer B from computer A and are unsuccessful. What is the problem?



(A) The IP address of R1 is on the wrong subnet.

(B) The default gateway setting of computer B is incorrect.

(C) The default gateway setting of computer A is incorrect.

(D) The cable type between the switches and routers is incorrect.

(E) The IP address of R2 is on the wrong subnet.

(F) The cable type between the computers and switches is incorrect.

36. You are testing communication to a router and have decided to connect your workstation to the Fast Ethernet port of the router. What type of cable would you use?

(A) Fiber

(B) Crossover

(C) Straight-through

(D) Thinnet

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37. You have a UTP cable that has been configured at both ends with the 568B standard. What type of cable is it?

(A) Straight-through

(B) Crossover

(C) Rollover

(D) Coax

38. You wish to create a crossover cable and have wired one end of the cable with the 568A standard, What standard should you use to wire the opposite end of the cable?

(A) 568B

(B) 568A

(C) 569B

(D) 569A